

Pavle Josić

Full-Stack Developer • Belgrade, Serbia

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PROFILE SUMMARY

Software and Information Engineering student focused on backend engineering, distributed systems, and application security (AppSec). Stack-agnostic developer experienced in building scalable server-side architectures, implementing defensive design patterns, and developing frontend interfaces with modern frameworks to deliver full-stack applications.

TECHNICAL ARSENAL

LANGUAGES: JavaScript • TypeScript • Python • Solidity • Java

FRAMEWORKS, LIBS & TOOLS: Next.js • React • Nest.js • Spring Boot • Express.js • Tailwind CSS • GSAP • Postman • Claude AI

BACKEND, OPS & DBS: Node.js • PostgreSQL • Prisma ORM • Docker • Redis

SELECTED WORKS

Microservices Restaurant Management System

[GitHub](#) →

Spring Boot • Java • PostgreSQL • Docker • Microservices • JWT

Built a distributed backend using microservices to decouple authentication from core business operations. Containerized PostgreSQL instances to enforce database-per-service isolation, integrated a centralized JWT validation layer, and implemented Role-Based Access Control (RBAC) for secure inter-service communication.

Decentralized Automated Market Maker (AMM)

[GitHub](#) →

Solidity • Smart Contracts • ERC20 • OpenZeppelin • DeFi Architecture

Built a Uniswap V2-style decentralized exchange deployed on Ethereum testnets. Implemented the Factory-Pair-Router pattern for dynamic liquidity pool deployments. Developed smart contracts to manage LP token minting via the Constant Product Formula and handled on-chain token swaps with automated protocol fees.

Tehnički Pregled Lazarević 1968

[Live Link](#) →

Next.js • GSAP • Tailwind CSS • Vercel

Developed a web platform for a technical inspection business with a focus on fast performance and clean UI. Implemented responsive animations using GSAP and Tailwind CSS, and optimized SEO performance to improve local search visibility and keyword rankings.

NNCraft: Deep Learning Framework from Scratch

[GitHub](#) →

Python • NumPy • Linear Algebra • Machine Learning

Built a dense neural network framework from scratch using NumPy to understand the underlying mathematics of machine learning. Implemented abstractions for custom network layers, designed matrix-based backpropagation pipelines, and added optimization algorithms including Adam, RMSprop, and regularizers.

EDUCATION & TIMELINE

Software and Information Engineering

Singidunum University (3rd year)

2023 – Present

Information Technology

Electrotechnical School "Rade Končar"

2019 – 2023